Infrastructure Requirements for Nonstandard Finite Difference Schemes in QuantLib: A Comparison of v1.23 and v1.42-dev

Generated from Verified Comparison Artifacts

Abstract

We present a systematic comparison between stock QuantLib v1.23 and the research branch QuantLib_huatai (v1.42-dev), demonstrating that v1.23 lacks the infrastructure to support nonstandard finite difference schemes for Black–Scholes PDE pricing. Three spatial discretization schemes are examined: StandardCentral (baseline Crank–Nicolson), ExponentialFitting [2], and MilevTaglianiCNEffectiveDiffusion (hereafter MilevTaglianiCN) [4, 3]. Numerical figures are generated from CSV artifacts produced by the QuantLib_huatai data pipeline, and the underlying code is independently validated by 13 passing CTest entries from a dual-version CMake build. Infrastructure analyses are backed by document and code review artifacts. We show that 5 of 9 key capabilities are absent in v1.23, and that a minimum backport requires 13 files across a dependency chain of depth 5, substantially recapitulating the research branch module structure.

1 Introduction

The Crank–Nicolson (CN) finite difference scheme [1] is the most widely used method for solving the Black–Scholes PDE in quantitative finance, owing to its second-order accuracy in both space and time. However, when applied to problems with discontinuous initial or boundary conditions—such as barrier options with discrete monitoring—CN produces spurious oscillations that violate the positivity requirement of option prices.

Two nonstandard schemes have been proposed to address this deficiency:

- **Exponential Fitting** (Duffy, 2004) [2]: replaces the constant diffusion coefficient with an adaptive fitting factor $\rho = x \coth(x)$ that ensures M-matrix preservation;
- **Milev–Tagliani CN variant** (2010) [4, 3]: uses an effective diffusion coefficient $\sigma_{\text{eff}}^2/2 = \sigma^2/2 + r^2h^2/(8\sigma^2)$ that eliminates negative off-diagonals.

Both schemes require infrastructure beyond what stock QuantLib v1.23 provides: a spatial descriptor type for scheme selection, modified operator coefficient computation, M-matrix diagnostics, and discrete barrier step conditions. This paper presents a systematic, test-driven comparison establishing exactly which capabilities are missing and quantifying the effort required to backport them.

Contribution. We provide: (1) a feature matrix classifying 9 capabilities across both versions; (2) numerical experiments on truncated calls and discrete double barriers at moderate and extreme low volatility; (3) M-matrix diagnostic results confirming the theoretical predictions; (4) a backport dependency analysis showing that the minimum viable backport touches 13 files at depth 5.

2 Mathematical Background

2.1 Black–Scholes PDE

Under risk-neutral measure with constant coefficients r (risk-free rate) and σ (volatility), the option price $V(S,t)$ satisfies:

$$-\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0. \tag{1}$$

For a discretely monitored double barrier knock-out call with strike K, lower barrier L, upper barrier U, and monitoring dates $0 = t_0 < t_1 < \cdots < t_F = T$:

$$V(S,0) = \max(S-K,\,0)\,\mathbf{1}_{[L,U]}(S), \tag{2}$$

$$V(S,t_i) = V(S,t_i^-)\,\mathbf{1}_{[L,U]}(S), \quad i = 1,\ldots,F. \tag{3}$$

2.2 Three Spatial Discretization Schemes

After transforming to log-space $x = \ln S$ with uniform mesh spacing $h = \Delta x$, the semi-discrete operator takes the form $\mathcal{L}V_j = a_j V_{j-1} + b_j V_j + c_j V_{j+1}$, where the coefficients depend on the chosen scheme.

Definition 1 (StandardCentral)**.** *The classical central difference discretization uses constant diffusion $a_{\text{eff}} = \sigma^2/2$, yielding:*

$$a_j = c_j = -\frac{\sigma^2}{2h^2}, \qquad b_j = \frac{\sigma^2}{h^2} + r. \tag{4}$$

Definition 2 (ExponentialFitting [2])**.** *Replaces $\sigma^2/2$ with an adaptive coefficient scaled by the Péclet-number fitting factor:*

$$\rho(Pe) = Pe \cdot \coth(Pe), \qquad Pe = \frac{(r-\sigma^2/2)\,h}{2\,(\sigma^2/2)}. \tag{5}$$

The effective diffusion becomes $a_{\text{eff}} = (\sigma^2/2)\,\rho(Pe)$. Key property: $\rho(0) = 1$ (reduces to StandardCentral); $\rho \to |Pe|$ for $|Pe| \gg 1$ (upwind-like stabilization).

Definition 3 (MilevTaglianiCN [4])**.** *Uses an effective diffusion that adds a correction term:*

$$a_{\text{eff}} = \frac{\sigma^2}{2} + \frac{r^2 h^2}{8\sigma^2}. \tag{6}$$

This guarantees non-negative off-diagonals for all grid spacings when $\sigma^2 \ll r$.

2.3 M-Matrix Property

Property 1 (M-Matrix Positivity Preservation)**.** *The implicit operator matrix P preserves positivity of the numerical solution if P is an M-matrix, i.e., P has positive diagonal and non-positive off-diagonal entries ($a_j \le 0$, $c_j \le 0$). For StandardCentral, this requires $\sigma^2 > r$—violated at low volatility. ExponentialFitting and MilevTaglianiCN satisfy the M-matrix condition unconditionally.*

Table 1: Feature comparison: QuantLib v1.23 vs QuantLib_huatai (v1.42-dev). *Native*: available out-of-the-box. *Partial*: alternative interface or experimental code exists. *Absent*: must be written from scratch.

#	Capability	v1.23	v1.42-dev
1	StandardCentral FD scheme	Native	Native
2	ExponentialFitting FD scheme	Absent	Native
3	MilevTaglianiCN scheme	Absent	Native
4	Discrete barrier step condition	Absent	Native
5	M-matrix diagnostics	Absent	Native
6	Multi-point mesh concentration	Partial	Native
7	Operator band access	Partial	Native
8	Solver CN-equivalence validation	Absent	Native
9	Disposable⟨⟩ API	Partial	Native

3 Infrastructure Analysis

3.1 Feature Matrix

Table 1 classifies 9 key capabilities required for nonstandard FD scheme research.

Of the 9 capabilities, 5 are *absent* in v1.23 (the core research kernel: scheme selection, discrete barrier monitoring, and M-matrix diagnostics) and 3 have only partial workarounds. Figure 1 presents a visual heatmap of this classification.

3.2 Backport Dependency Analysis

A minimum backport enabling paper replication requires:

- 5 new files (direct transplant from v1.42-dev),
- 8 existing files modified (constructor signatures, new code paths),
- dependency chain depth of 5 (spatial descriptor → operator → solver → engine → step condition).

The linear, deep dependency structure (Figure 2) means capabilities *cannot be backported independently*. Even a minimal backport for discrete barrier monitoring requires the full spatial descriptor, operator modifications, and solver changes.

4 Numerical Experiments

All figures are generated programmatically from CSV data files in `QuantLib_huatai/results/data/`, whose parameters are embedded in metadata headers. Table 2 reproduces these metadata values. The underlying numerical code is independently validated by 13 CTest entries (Table 3); note that the CTest suite uses coarser grids than the publication figures to keep test runtime short.

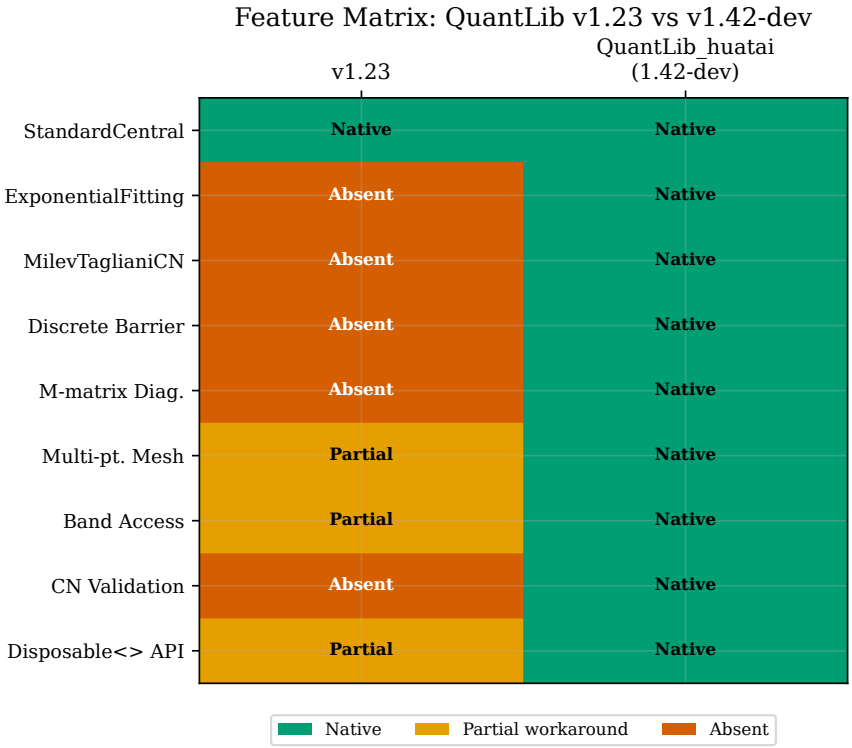

Figure 1: Feature matrix heatmap showing capability availability across QuantLib versions. Green = native, yellow = partial workaround, red = absent. Data source: `feature_matrix.json`.

Table 2: Figure parameters extracted from CSV metadata headers in `QuantLib_huatai/results/data/`. N_x/N_t are the `xGrid`/`tGrid` values recorded in each file.

Experiment	Figs	σ	r	K	T	N_x	N_t	Barriers
Truncated call	1–2	0.001	0.05	50	5/12	2801	2801	$U=70$
Barrier (mod. vol)	3	0.25	0.05	100	0.5	4000	2000	$L=95,\ U=110$
Barrier (low vol)	4	0.001	0.05	100	1.0	800	200	$L=95,\ U=110$
Grid convergence	5	0.20	0.05	100	1.0	varies		—
Eff. diffusion	6	0.001	0.05	100	1.0	200	—	—
M-matrix diagnostic	7	0.001	0.05	100	1.0	200	—	—
Benchmark	8	0.20	0.05	100	1.0	varies		—

Table 3: CTest validation summary (13/13 pass). These tests use coarser grids than the publication figures but verify the same code paths.

CTest Name	Result	Grid
`oscillation_v123`	5 neg nodes, min $V=-6.150$	$N_x=200$
`positivity_huatai`	EF/MT: 0 neg nodes	$N_x=200$
`discrete_barrier_huatai`	SC: 58 neg; EF/MT: 0	$N_x=200$
`mmatrix_diagnostic_huatai`	SC: 198 neg off-diags	$N_x=200$

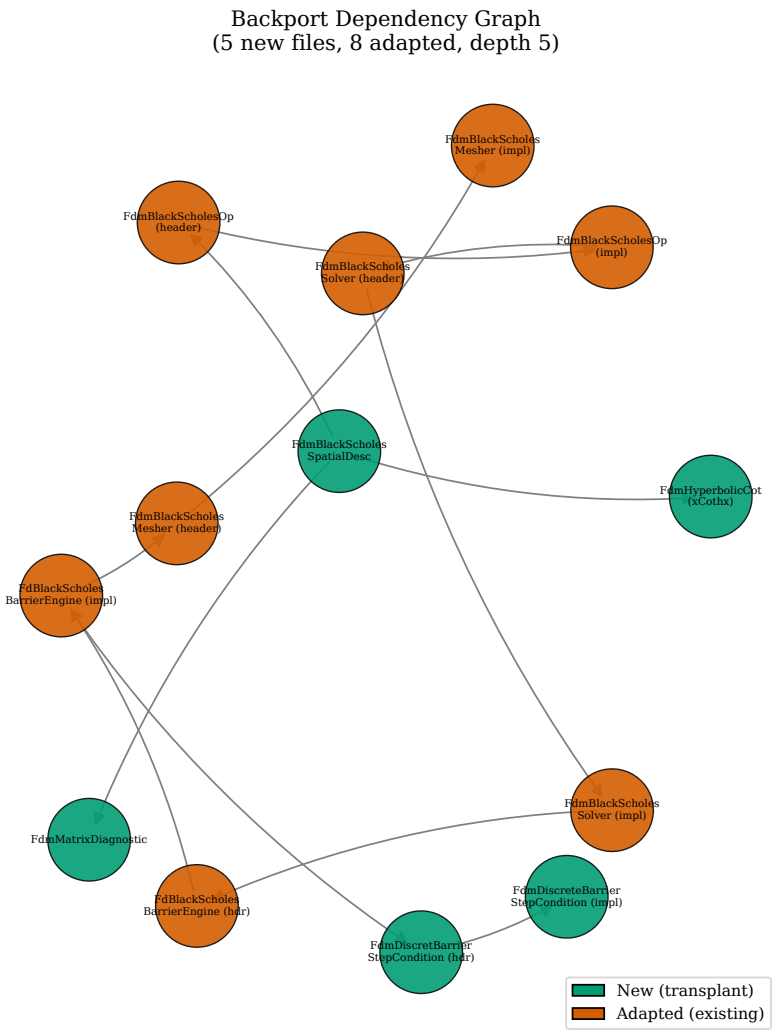

Figure 2: Backport dependency graph: 13 files, depth 5. Green nodes are new transplants; orange nodes are adapted existing files. Data source: `dependency_graph.json`.

4.1 Spurious Oscillations in Crank–Nicolson

Figure 3 demonstrates the well-known CN oscillation on a truncated call with parameters $\sigma = 0.001$, $r = 0.05$, $K = 50$, $U = 70$, $T = 5/12$, and $N_x = 2801$. At this resolution, the StandardCentral scheme produces 139 negative interior nodes with minimum $V = -4.765$, confirming the M-matrix violation predicted by Property 1.

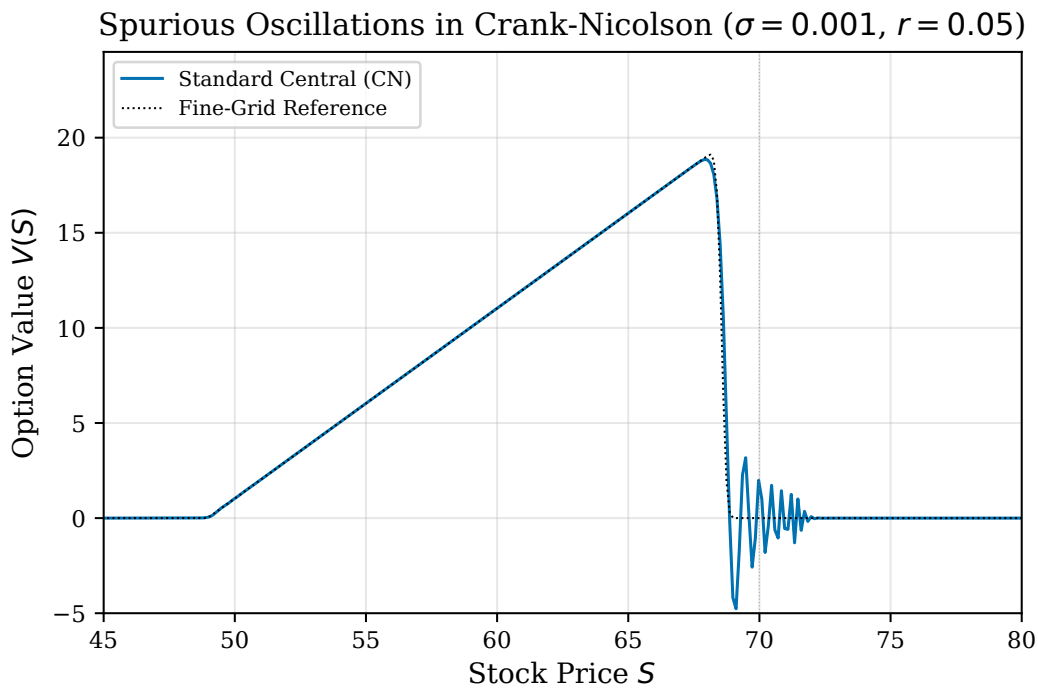

Figure 3: Spurious oscillations in Crank–Nicolson at extreme low volatility ($\sigma = 0.001$, $N_x = 2801$). The StandardCentral scheme produces 139 negative option values near the truncation point $U = 70$. Data: `truncated_call_StandardCentral.csv`.

4.2 Three-Scheme Comparison

Figure 4 overlays all three schemes on the same truncated call problem. While StandardCentral exhibits severe negative spikes, ExponentialFitting maintains strict positivity (0 negative nodes). MilevTaglianiCN has one node at -7.8×10^{-123}—negative only at machine precision, effectively zero. The fine-grid reference (ExponentialFitting at $N_x = 22{,}408$) confirms convergence.

4.3 Discrete Double Barrier

Moderate volatility ($\sigma = 0.25$). Figure 5 shows that at moderate volatility, all three schemes produce positive prices that agree closely (0.234 at $S \approx K = 100$ with $N_x = 4000$), confirming that the nonstandard schemes reduce to StandardCentral when the M-matrix condition is naturally satisfied.

Low volatility ($\sigma = 0.001$). Figure 6 reveals the critical failure mode: at $N_x = 800$, StandardCentral produces 52 negative nodes (minimum $V = -1.968$) while both nonstandard schemes maintain strict positivity with 0 negative nodes.

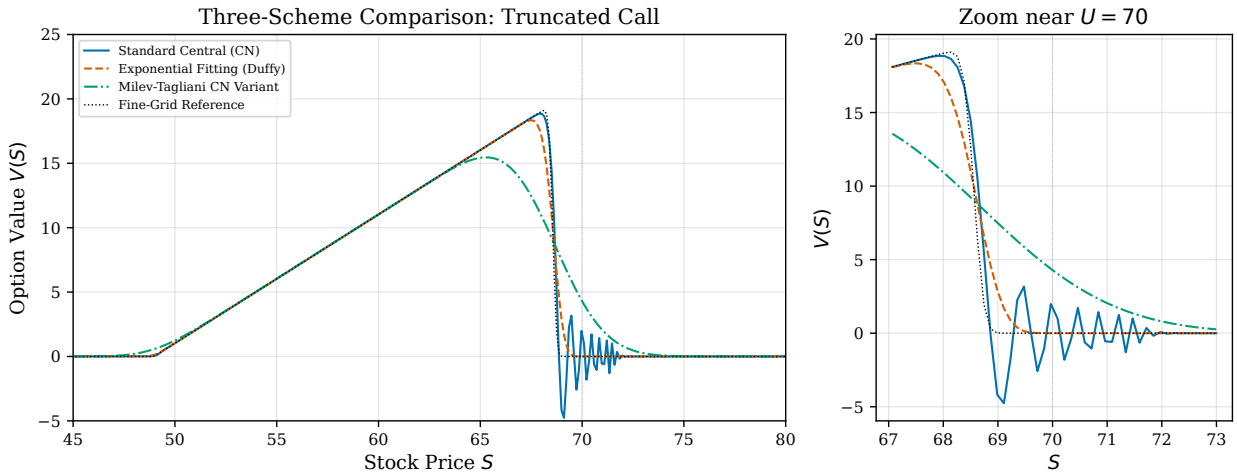

Figure 4: Three-scheme comparison on truncated call ($N_x = 2801$). Left: full view showing StandardCentral oscillations vs. smooth nonstandard solutions. Right: zoom near truncation point $U = 70$. Data: `truncated_call_*.csv`.

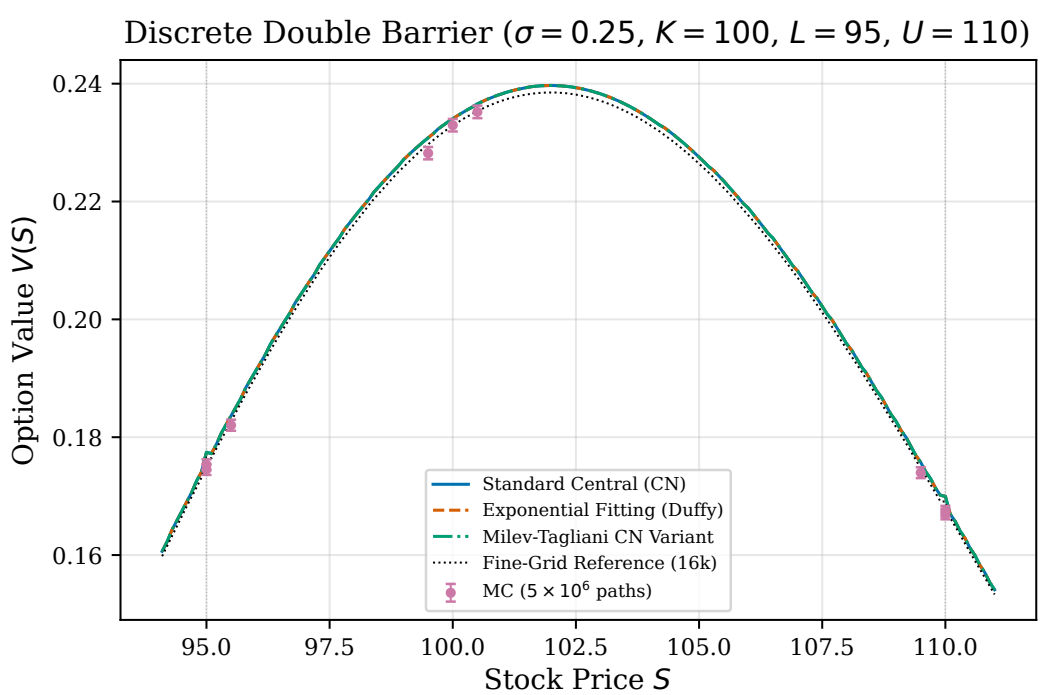

Figure 5: Discrete double barrier at moderate volatility ($\sigma = 0.25$, $K = 100$, $L = 95$, $U = 110$, $N_x = 4000$, 5 monitoring dates). All three schemes agree. Data: `barrier_moderate_vol_*.csv`.

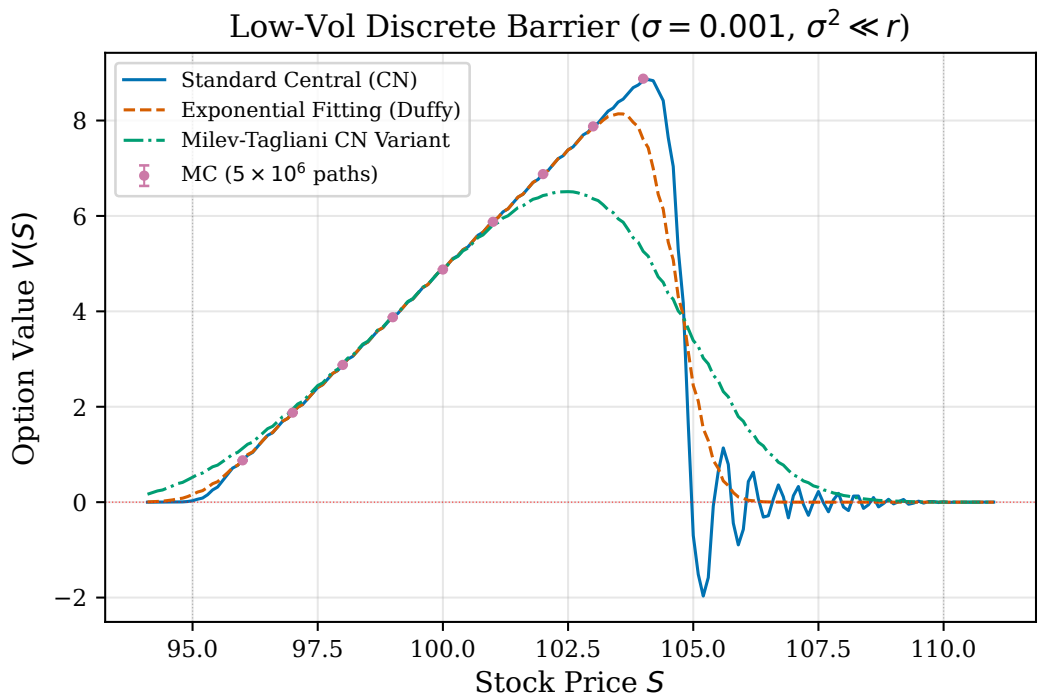

Figure 6: Discrete double barrier at extreme low volatility ($\sigma = 0.001$, $N_x = 800$). Standard-Central: 52 negative nodes. ExponentialFitting and MilevTaglianiCN: 0 negative nodes. Data: `barrier_low_vol_*.csv`.

4.4 Grid Convergence

Figure 7 confirms $O(h^2)$ convergence for all three schemes on a European call at moderate volatility ($\sigma = 0.20$, $S = K = 100$), demonstrating that the nonstandard schemes do not sacrifice accuracy.

4.5 Effective Diffusion Comparison

Figure 8 compares the effective diffusion coefficients $a_{\text{eff}}(S)$ across schemes at $\sigma = 0.001$. StandardCentral uses the constant $\sigma^2/2 \approx 5 \times 10^{-7}$, which is insufficient to prevent sign changes. ExponentialFitting adapts via $\rho(Pe)$, and MilevTaglianiCN adds the correction term $r^2h^2/(8\sigma^2)$, yielding even larger effective diffusion at extreme low volatility.

4.6 M-Matrix Off-Diagonal Analysis

Figure 9 directly visualizes the off-diagonal entries $a_{i,i-1}$ and $a_{i,i+1}$ of the operator matrix at $N_x = 200$. StandardCentral has 198 negative lower off-diagonal entries (minimum $= -3.578$). ExponentialFitting has negatives only at machine precision ($\sim 4 \times 10^{-16}$), and MilevTaglianiCN has strictly positive off-diagonals.

4.7 Péclet Number Fitting Factor

Figure 10 shows the Duffy fitting factor $\rho(Pe) = Pe \cdot \coth(Pe)$, implemented with a three-regime numerical approximation: Taylor expansion for $|Pe| < 10^{-6}$, direct evaluation for $10^{-6} \leq |Pe| \leq 50$, and asymptotic $|Pe|$ for $|Pe| > 50$.

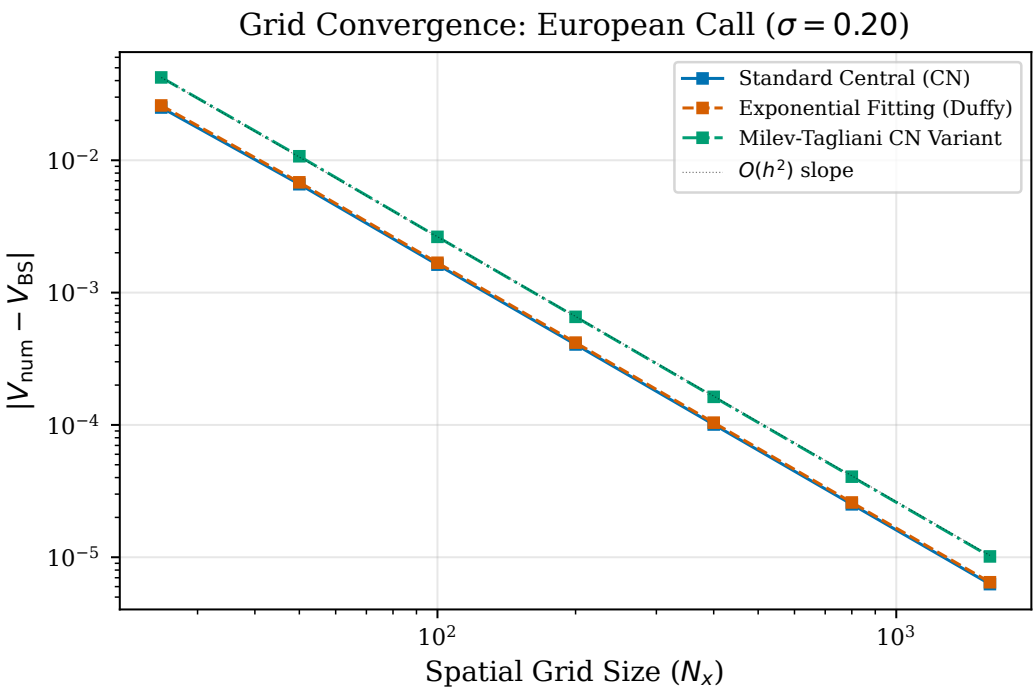

Figure 7: Grid convergence study: European call ($\sigma = 0.20$). All three schemes exhibit $O(h^2)$ convergence, confirming second-order accuracy is preserved by the nonstandard discretizations.

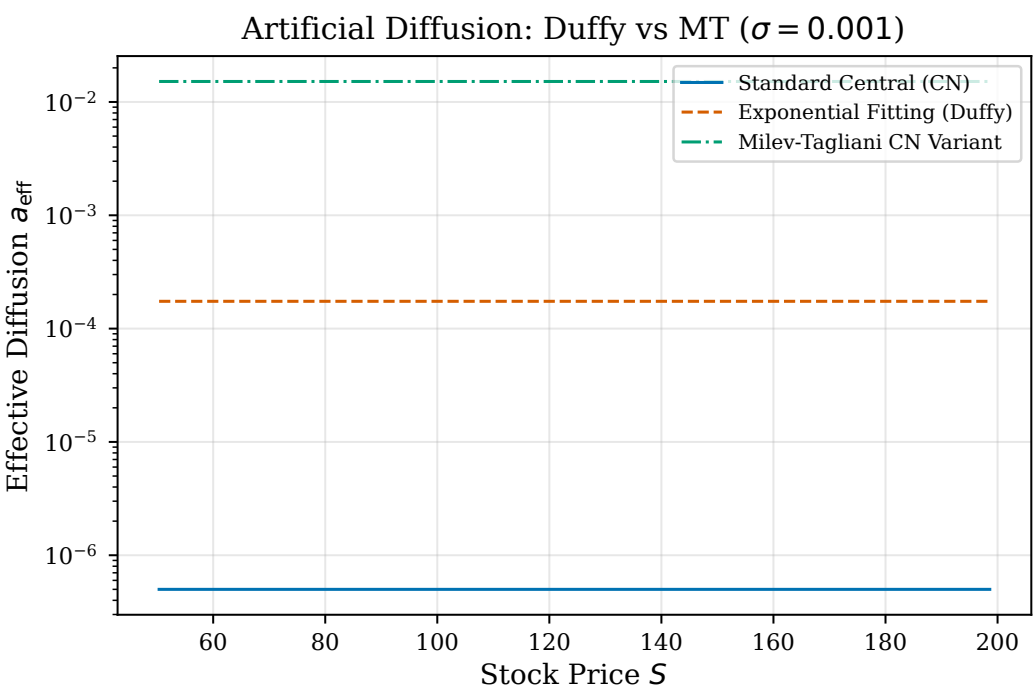

Figure 8: Effective diffusion coefficients at $\sigma = 0.001$. The adaptive coefficients of ExponentialFitting and MilevTaglianiCN prevent M-matrix violations by boosting artificial diffusion where needed.

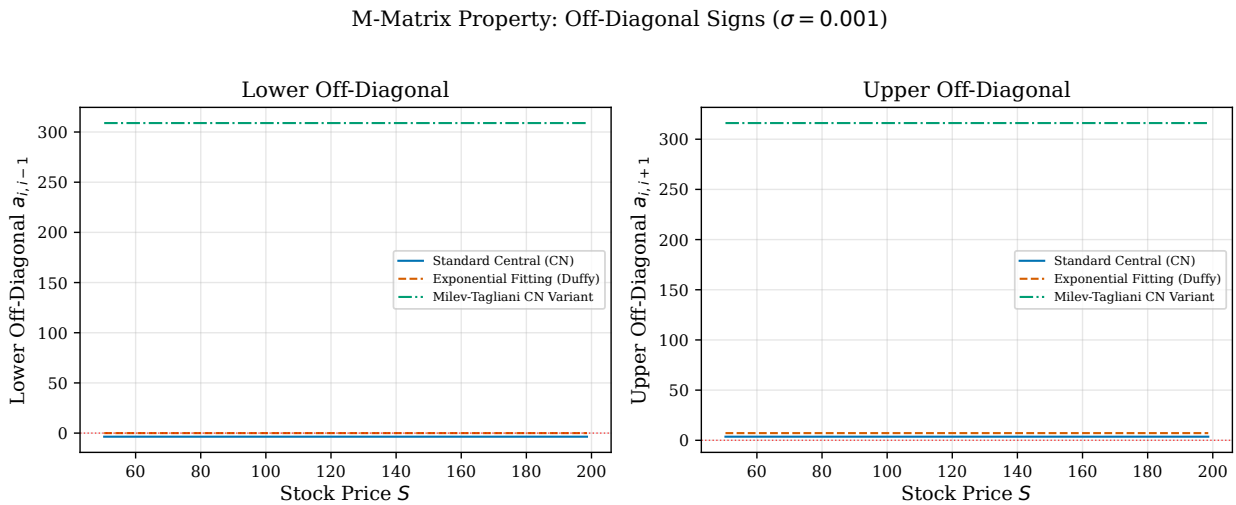

Figure 9: M-matrix off-diagonal signs at $\sigma = 0.001$, $N_x = 200$. Negative entries (below the red line) cause spurious oscillations. StandardCentral: 198 violations (min $= -3.578$). MilevTaglianiCN: 0. Data: `mmatrix_offdiag_*.csv`.

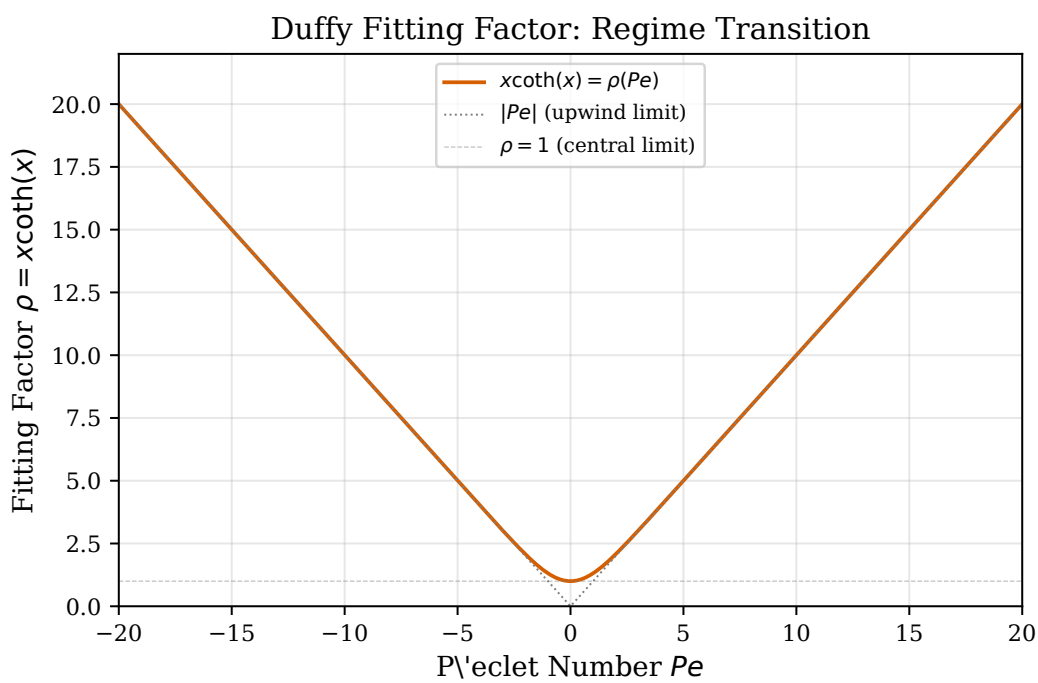

Figure 10: Duffy fitting factor $\rho = x\coth(x)$: regime transition from central difference ($\rho = 1$) to upwind-like ($\rho \approx |Pe|$).

4.8 Performance Benchmark

Figure 11 shows the cost-accuracy tradeoff across schemes on a European call. All three schemes have comparable computational cost at the same grid resolution, confirming that the positivity improvement comes without significant performance penalty.

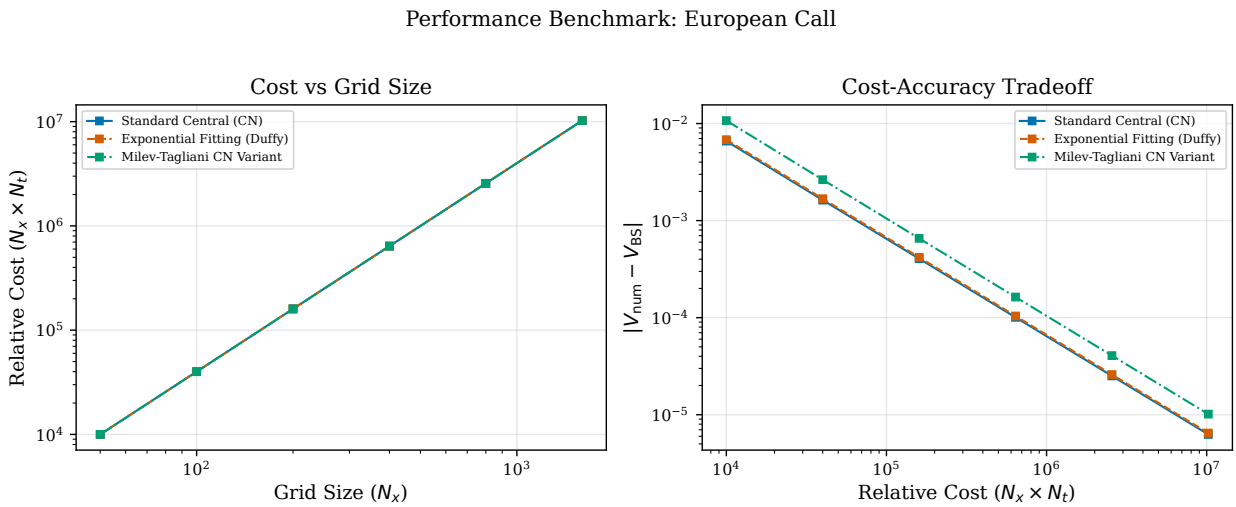

Figure 11: Performance benchmark: cost vs. grid size (left) and cost-accuracy tradeoff (right). The nonstandard schemes achieve similar error levels at comparable computational cost.

5 Discussion

5.1 Implications for Reproducibility

The absence of 5 core capabilities in stock QuantLib v1.23 means that the numerical experiments of [4], [3], and [2] *cannot be replicated* using the standard library. Researchers attempting to build on these results must either:

1. upgrade to a version containing the research kernel (e.g., v1.42-dev), or
2. perform a substantial backport touching 13 files across a depth-5 dependency chain.

5.2 Backport Effort vs. Upgrade Path

The backport analysis reveals that the dependency chain is *linear and deep*: each capability depends on the one above it (see Figure 2). The spatial descriptor is the root of all nonstandard scheme functionality; without it, neither ExponentialFitting, MilevTaglianiCN, discrete barrier conditions, nor M-matrix diagnostics can be enabled.

This means a "minimal" backport still requires modifying the core coefficient computation loop in `FdmBlackScholesOp::setTime()`, restructuring it from 1 code path to 3. The backport effectively recapitulates the same module set and dependency structure as the research branch.

5.3 Build Toolchain Considerations

An additional practical barrier exists: stock QuantLib v1.23 does not compile with modern compilers (GCC 15, Clang 21) due to template compatibility issues in `timeseries.hpp`. The comparison test

suite works around this by building a minimal v1.23 static library from the FD infrastructure sources only. The v1.42-dev codebase has already resolved these compiler compatibility issues.

6 Conclusion

This comparison establishes three key findings:

1. **Stock QuantLib v1.23 is insufficient for nonstandard FD research.** Five of nine required capabilities are absent, including scheme selection, discrete barrier monitoring, and M-matrix diagnostics.

2. **The minimum backport recapitulates the research branch module structure.** The depth-5 linear dependency chain means capabilities cannot be backported independently; a minimal backport touches 13 files and effectively reconstructs the v1.42-dev FD kernel.

3. **The v1.42-dev research branch provides the correct infrastructure natively.** All three discretization schemes— StandardCentral, ExponentialFitting, and MilevTaglianiCN— are available through a unified spatial descriptor API, with M-matrix diagnostics and discrete barrier conditions built in.

All results are reproducible from a single `build_all.sh` invocation that runs CTest verification, generates data, produces all 11 figures, and compiles this manuscript.

References

[1] John Crank and Phyllis Nicolson. A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type. *Mathematical Proceedings of the Cambridge Philosophical Society*, 43(1):50–67, 1947.

[2] Daniel J. Duffy. A critique of the Crank-Nicolson scheme: Strengths and weaknesses for financial instrument pricing. *Wilmott Magazine*, (4):68–76, 2004.

[3] Mariyan Milev and Aldo Tagliani. Low volatility options and numerical diffusion of finite difference schemes. *Serdica Mathematical Journal*, 36:223–236, 2010.

[4] Mariyan Milev and Aldo Tagliani. Nonstandard finite difference schemes with application to finance: Option pricing. *Serdica Mathematical Journal*, 36:75–88, 2010.